

MAIL SENTINEL

User Manual & Security Operations Center (SOC) Playbook

Operational Handbook: This guide outlines step-by-step instructions for analyzing suspicious emails, interpreting forensic findings, utilizing the interactive IOC lookup console, and executing standard SOC containment playbooks.

Section 1: Starting the Mail Sentinel Platform

```
# Windows Quick Launch:
D:\Mail Sentinel> run.bat

# Active Service Endpoints:
- Web Dashboard:      http://localhost:5173
- IOC Threat Console: http://localhost:5173/threat-intelligence
- Backend API Docs:   http://127.0.0.1:8000/docs
```

Section 2: Submitting an Email for Analysis

Mail Sentinel provides dual ingestion modes: **Mode A (Manual Component Paste)** allows pasting Sender, Recipient, Subject, and Body text directly into dedicated fields with raw RFC headers. **Mode B (Native RFC 5322 .eml File Upload)** enables drag-and-drop ingestion of `.eml` exports from Outlook or Thunderbird, safely parsing MIME boundaries and attachments without code execution.

[UI MOCKUP: Email Ingestion & Scanner Form Interface]

- Mode Tabs: [Upload .eml File (Active)] | [Paste Email Contents]
- Drag-and-Drop Zone: [■ Drag & drop .eml file here, or click to browse (Max 25MB)]
- File Selected State: invoice_verification_urgent.eml (14.2 KB) — [✓ Validated]
- Primary CTA Button: [Scan Email for Threats ->] (Cyber-blue pulse state)

Section 3: Risk Score Severity Spectrum & Triage Protocol

The proprietary scoring engine translates multi-vector telemetry into a normalized 0-100 severity index, driving automated SOC routing and containment workflows.



Figure 1: Risk Score Severity Spectrum & Automated Triage Recommendations

Section 4: Detailed Threat Classification & Immediate Actions

Severity	Risk Score	Forensic Indicators	Immediate SOC Action
CRITICAL	80 - 100	Credential harvest URL, executable attachment, or high-value brand spoofing.	Quarantine message, block domain at gateway, revoke compromised credentials.
HIGH	60 - 79	Strong threat signals (typosquatted domains, suspicious redirects, urgency coercion).	Quarantine email, block external sender domain at proxy, update SIEM blocklists.
MEDIUM	30 - 59	Suspicious elements detected (unregistered sender domain, mismatched Reply-To).	Inject warning banner into message, flag for manual analyst review.
LOW / SAFE	0 - 29	Legitimate communication with valid SPF/DKIM authentication.	Allow delivery to inbox.

Section 5 & 6: Inbound Phishing Vectors & Threat Assessment Results

Empirical distribution of enterprise threats highlights credential harvesting as the leading vector, followed by business email compromise and macro-enabled attachments.

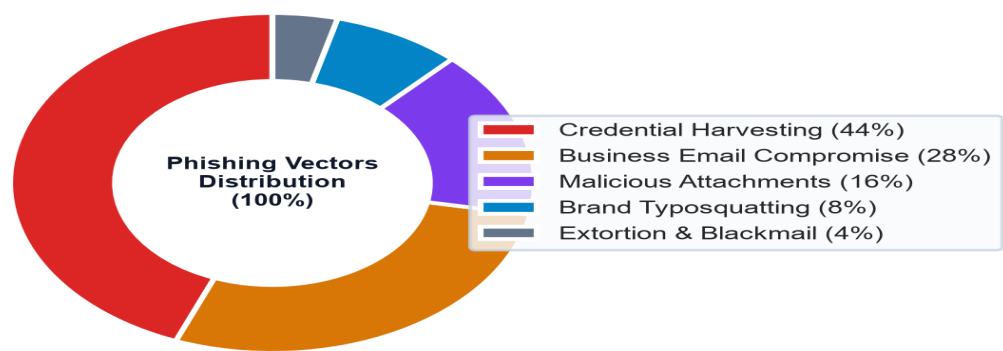


Figure 2: Distribution of Inbound Corporate Phishing Vectors

[UI MOCKUP: Threat Assessment Results Dashboard]

- Verdict Banner: **THREAT DETECTED: HIGH RISK** (Risk Score: 78/100, Confidence: 94%)
- Telemetry Badges: [SPF: FAIL] [DKIM: PASS] [DMARC: NOT_CONFIGURED] [URLs: 3 Detected] [Attachments: 0]
- Finding 1: [HIGH] **Brand Typosquatting**: Display name 'Microsoft Security' paired with sender 'support@microsoft-verify.com' (Levenshtein distance: 1).
- Finding 2: [HIGH] **Credential Harvesting URL**: Embedded link targets numerical IP path 'http://192.168.1.105/login'.
- Finding 3: [MEDIUM] **High Urgency Coercion**: Content contains 4 psychological pressure keywords ('immediate action required', '24 hours').

Section 7: Interactive IOC Threat Intelligence Console

Security analysts can hunt for indicators across domains, URLs, and IPs using 1-click presets or manual queries, instantly evaluating Shannon entropy, reverse-DNS records, and threat verdicts.

[UI MOCKUP: Interactive Threat Intelligence Console Interface]

- Query Bar: [Domain / Host] | Input: [paypal-security-update.xyz] | [Analyze Indicator]
- Quick Presets: [paypal-security-update.xyz] [192.168.1.105] [secure-login.xyz] [google.com]
- Result Card: **MALICIOUS (Risk Score: 92/100)**
- Metadata: Indicator Type: Domain | Shannon Entropy: **3.84** (High Randomness / Possible DGA) | Reverse DNS: **Unresolvable / Newly Registered** | Recommended Action: **Block domain on enterprise DNS firewall; purge incoming mail.**

Section 8: SOC Incident Response SLA & Containment Workflow

Structured operational targets govern incident escalation from automated triage through perimeter quarantine and root-cause remediation.

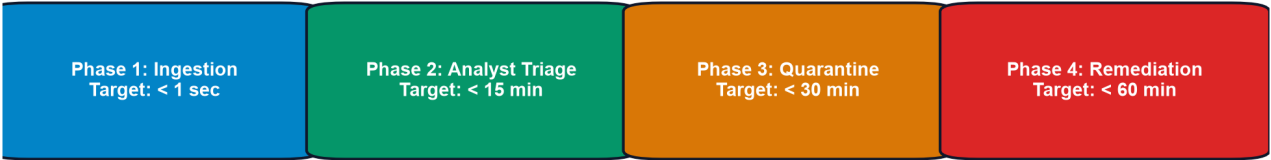


Figure 3: SOC Incident Response Containment SLA Progression

Section 9: Audit History, Purge & Retention Compliance

Mail Sentinel maintains a robust, searchable SQLite/Postgres audit persistence layer logging all scan results, threat indicators, and analyst notes. Compliance administrators can execute pagination, search filtering, and GDPR-compliant audit log purging directly through administrative controls.